remel

Part of Thermo Fisher Scientific

SAFETY DATA SHEET

Creation Date 11-May-2009 **Revision Date** 11-May-2009 **Revision Number** *1*

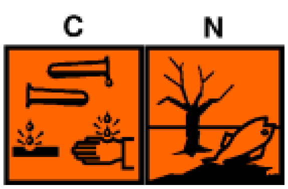
1. IDENTIFICATION OF THE SUBSTANCE/PREPARATION AND OF THE COMPANY/UNDERTAKING

Product Name **Ammonium Hydroxide**
Cat No. **R21195**
Synonyms No information available.
Recommended Use Laboratory chemicals

Company Remel
12076 Santa Fe Drive
Lenexa, KS 66215 United States
Telephone: 1-800-255-6730
Fax:1-800-621-8251

Emergency Telephone Number INFOTRAC - 24 Hour Number: 1-800-535-5053
Outside of the United States, call 24 Hour Number: 001-352-323-3500 (Call Collect)

2. HAZARDS IDENTIFICATION



R -phrase(s)
R34 - Causes burns
R50 - Very toxic to aquatic organisms

3. COMPOSITION/INFORMATION ON INGREDIENTS

Haz/Non-haz

Component	Weight %	EC No.	Classification
Ammonium hydroxide 1336-21-6	100	EEC No. 215-647-6	C;R34 N;R50

For the full text of the R phrases mentioned in this Section, see Section 16

4. FIRST AID MEASURES

Eye Contact	Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Immediate medical attention is required.
Skin Contact	Wash off immediately with plenty of water for at least 15 minutes. Immediate medical attention is required.
Ingestion	Do not induce vomiting. Call a physician or Poison Control Center immediately.
Inhalation	Move to fresh air. If breathing is difficult, give oxygen. Do not use mouth-to-mouth resuscitation if victim ingested or inhaled the substance; induce artificial respiration with a respiratory medical device. Immediate medical attention is required.
Notes to Physician	Treat symptomatically.

5. FIRE-FIGHTING MEASURES

Suitable Extinguishing Media
Use extinguishing measures that are appropriate to local circumstances and the surrounding environment

Extinguishing media which must not be used for safety reasons
No information available.

Specific Hazards Arising from the Chemical
Keep product and empty container away from heat and sources of ignition. Corrosive Material.

Protective Equipment and Precautions for Firefighters
As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear. Thermal decomposition can lead to release of irritating gases and vapors.

Flash Point	Not applicable
Method	No information available.
Autoignition Temperature	No information available.
Explosion Limits	
Lower	16 Vol %
Upper	27 Vol %

6. ACCIDENTAL RELEASE MEASURES

Personal Precautions	Ensure adequate ventilation. Use personal protective equipment. Keep people away from and upwind of spill/leak. Evacuate personnel to safe areas.
Environmental Precautions	Should not be released into the environment.
Methods for Containment and Clean Up	Soak up with inert absorbent material Keep in suitable and closed containers for disposal

7. HANDLING AND STORAGE

Handling Use only under a chemical fume hood. Wear personal protective equipment. Do not get in eyes, on skin, or on clothing. Do not breathe vapors/dust. Do not ingest.

Storage Keep containers tightly closed in a dry, cool and well-ventilated place. Corrosives area.

Specific use(s) No information available.

8. EXPOSURE CONTROLS / PERSONAL PROTECTION

Exposure limits

Component	Italy	Portugal	The Netherlands	Finland	Austria
Ammonium hydroxide				TWA: 14 mg/m^3 TWA: 20 ppm STEL: 36 mg/m^3 STEL: 50 ppm	

Occupational exposure controls

Engineering Measures Use only under a chemical fume hood Ensure adequate ventilation, especially in confined areas Ensure that eyewash stations and safety showers are close to the workstation location

Personal Protective Equipment

Respiratory Protection Follow the OSHA respirator regulations found in 29 CFR 1910.134 or European Standard EN 149. Use a NIOSH/MSHA or European Standard EN 149 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.

Eye Protection Tightly fitting safety goggles

Skin and body protection Long sleeved clothing

Hand Protection Protective gloves

Hygiene Measures Handle in accordance with good industrial hygiene and safety practice

Environmental exposure controls No information available.

9. PHYSICAL AND CHEMICAL PROPERTIES

Physical State Liquid
Appearance Clear, Colorless
pH 13.5
Vapor Pressure 115 mmHg @ 20°C
Boiling Point/Range 360°C / 680°F
Melting Point/Range -72°C / -97.6°F
Flash Point Not applicable
Explosion Limits
Lower 16 Vol %
Upper 27 Vol %
Specific Gravity 0.9

9. PHYSICAL AND CHEMICAL PROPERTIES

Molecular Formula	NH4OH
Molecular Weight	35.05

10. STABILITY AND REACTIVITY

Stability	Stable under normal conditions.
Conditions to Avoid	Incompatible products. Heat, flames and sparks.
Incompatible Materials	Oxidizing agents, Acids, Acid anhydrides, Acid chlorides, Halogens, Metals
Hazardous Decomposition Products	Carbon monoxide (CO), Carbon dioxide (CO_2)
Hazardous Polymerization	Hazardous polymerization does not occur
Hazardous Reactions .	None under normal processing.

11. TOXICOLOGICAL INFORMATION

Acute Toxicity

Component Information

Component	LD50 Oral	LD50 Dermal	LC50 Inhalation
Ammonium hydroxide	350 mg/kg (Rat)		

Chronic Toxicity

Carcinogenicity	There are no known carcinogenic chemicals in this product
Sensitization	No information available.
Neurological Effects	No information available.
Mutagenic Effects	Mutatagenic effects have occured in microorganisms.
Reproductive Effects	No information available.
Developmental Effects	No information available.
Target Organs	Skin, Respiratory system, Eyes, Gastrointestinal tract (GI), Kidney.
Other Adverse Effects	The toxicological properties have not been fully investigated.. See actual entry in RTECS for complete information.

12. ECOLOGICAL INFORMATION

Ecotoxicity

. Very toxic to aquatic organisms.

Component	Freshwater Algae	Freshwater Fish	Microtox	Water Flea
Ammonium hydroxide		LC50= 8.2 mg/L Pimephales promelas 96 h		EC50 = 0.66 mg/L 48 h

Persistence and Degradability No information available

Bioaccumulative Potential No information available.

Mobility No information available.

13. DISPOSAL CONSIDERATIONS

Waste from Residues / Unused Products Chemical waste generators must determine whether a discarded chemical is classified as a hazardous waste. Chemical waste generators must also consult local, regional, and national hazardous waste regulations to ensure complete and accurate classification

Contaminated Packaging Empty containers should be taken for local recycling, recovery or waste disposal

14. TRANSPORT INFORMATION

IMDG/IMO

UN-No UN2672
Hazard Class 8
Packing Group III
Proper Shipping Name AMMONIA SOLUTION

ADR

UN-No UN2672
Hazard Class 8
Packing Group III
Proper Shipping Name AMMONIA SOLUTION

IATA

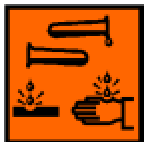
UN-No UN2672
Hazard Class 8
Packing Group III
Proper Shipping Name AMMONIA SOLUTION

15. REGULATORY INFORMATION

Labelling

Symbol(s) C - Corrosive

C



N

R -phrase(s)
R34 - Causes burns
R50 - Very toxic to aquatic organisms

S -phrase(s)
S26 - In case of contact with eyes, rinse immediately with plenty of water and seek medical advice
S45 - In case of accident or if you feel unwell, seek medical advice immediately (show the label where possible)
S61 - Avoid release to the environment. Refer to special instructions/safety data sheets
S36/37/39 - Wear suitable protective clothing, gloves and eye/face protection

International Inventories

Component	EINECS	ELINCS	NLP	TSCA	DSL	NDSL	PICCS	ENCS	CHINA	AICS	KECL
Ammonium hydroxide	215-647-6	-		Present	X	-	X	X	X	X	KE-01688 X

16. OTHER INFORMATION

Text of R phrases mentioned in Section 2-3
R34 - Causes burns
R50 - Very toxic to aquatic organisms

Prepared By Regulatory Affairs

Creation Date 11-May-2009

Revision Date 11-May-2009

Revision Summary "***", and red text indicates revision

Restrictions on use No information available.

Training advice No information available.

Literary reference No information available.

Disclaimer

The information provided on this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guide for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered as a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other material or in any process, unless specified in the text.

End of Safety Data Sheet